

Subjective Quality Assessment of User-Generated 360° Videos

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ABSTRACT

In this poster, we establish one of the largest virtual reality (VR) video database, containing 502 user-generated videos with rich content and commingled *authentic* distortions (often localized in space and time). We capture viewing behaviors (*i.e.*, scanpaths) of 139 users, and collect their opinion scores of perceived quality under four different viewing conditions (two starting points $\times$ two exploration times). We provide a thorough statistical analysis of recorded data, resulting in several interesting observations, which reveal how viewing conditions affect human behaviors and perceived quality. The database is available at <https://github.com/Yao-Yiru/VR-Video-Database>.

Index Terms: 360° videos—Scanpaths—Perceived quality

1 INTRODUCTION

As virtual reality (VR) acquisition and display systems become widely accessible, people are getting used to capturing, editing, and interacting with VR content, which is evidenced by the accelerated proliferation of 360° videos uploaded to popular video sharing and social media platforms (*e.g.*, Bilibili and Youtube). A practical issue arising from user-generated 360° videos is that they come with complex visual artifacts (*i.e.*, the so-called *authentic* distortions) from acquisition to processing stages [2]. Thus, understanding how people perceive 360° video distortions in virtual environments is central to many VR-enabled video applications.

Although several subjective quality studies on 360° videos have been conducted, they may suffer from three limitations. First, most of the resulting databases contain *synthetic* distortions only, which is an oversimplification of the real-world situation. Second, without the constraint of viewing conditions (*e.g.*, starting points or exploration durations), some distortions in a 360° will be invisible (or only visible) to some users, and thus it is reasonable for those users to rate the perceptual quality as perfect (or poor), leading to rating inconsistency. Third, given a fixed human labeling budget, the number of reference 360° videos in the databases is limited to a few dozen (if not fewer). As such, existing databases fail to sufficiently represent real-world videos with diverse content and distortion.

In view of the above limitations of existing 360° video databases, many important problems remain under-explored for understanding the perceived quality of 360° videos. For example, how consistent are human behaviors under the same viewing condition? How are human behaviors affected by viewing conditions? How does perceived quality change with viewing conditions?

In an attempt to answer these questions, we establish a large 360° video database, which contains 502 360° video sequences to span diverse video content. The videos exhibit a wide range of complex authentic distortions, covering the full quality spectrum. To assess the perceived quality of a 360° video as a function of viewing

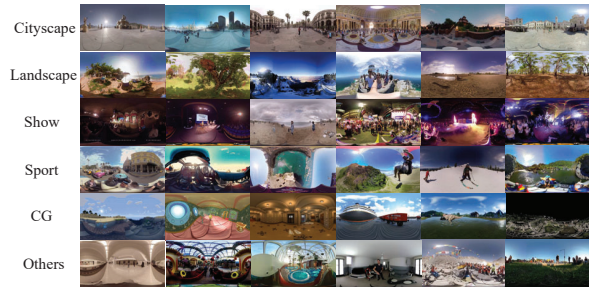


Figure 1: Thumbnails of user-generated 360° video sequences in the proposed dataset.

conditions, we invite subjects to watch each video from different starting points and time durations. By doing so, a total of 40,268 perceptual opinion scores together with the scanpaths (as viewing behaviors) from 139 users are recorded. We then provide an in-depth analysis of our data, investigating the impact of viewing conditions on viewing behaviors and perceived quality.

2 PSYCHOPHYSICAL EXPERIMENT

2.1 Visual Stimuli

Our database contains 502 unique user-generated 360° video sequences, with frame rate ranging from 20 to 60 frames per second (fps) and resolution ranging from $1,280 \times 720$ to $5,120 \times 2,560$ pixels. All videos are downloaded from the Internet, carrying Creative Commons licenses. Each video is cropped to a duration of about 15 seconds and stored in the equirectangular projection (ERP) format without further compression. The videos are selected to span a diversity of scenes that are ideal for VR shooting: *Cityscape*, *Landscape*, *Shows*, *Sports*, *CG*, and *Others* (see Fig. 1). We primarily focus on authentic distortions, manifesting themselves as complex mixtures of multiple visual artifacts that arise during 360° video creation [4].

2.2 Viewing Conditions

We use an HTC Eye Pro to display 360° videos, which provides an FoV of 110° and a binocular resolution of $2,880 \times 1,600$ pixels. Subjects are asked to seat on a swivel chair using the head-mounted display (HMD) to watch videos. Eye data (EM) and head data (HM) can be collected by a built-in Tobii Pro eye-tracking system with a sampling rate of $2 \times$ fps.

An important consideration in our psychophysical experiment is that we vary two viewing conditions: the starting point and the exploration time. We intentionally choose *Starting Point I* to give users a *poor* initial viewing experience. From the Starting Point I, the initial viewports contain localized distortions or intensive spatiotemporal information. On the contrary, *Starting Point II* is selected to encourage a *good* initial viewing experience, and is at least 120° (in longitude) apart from Starting Point I. From Starting Point II, people can barely see visual distortions from the initial viewport. We also set two exploration times: one spanning the entire duration (*i.e.*, 15 seconds) and the other set to the half of the

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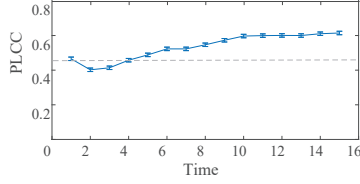


Figure 2: Viewing behavior consistency in terms of PLCC between heatmaps from Starting Point I and Starting Point II, averaged over all 360° videos. The dashed line represents the initial PLCC.

		Starting Point I	Starting Point II
mTC	7-second	0.394 (± 0.046)	0.395 (± 0.041)
	15-second	0.289 (± 0.038)	0.286 (± 0.034)
SRM	7-second	72.997 (± 7.221)	74.702 (± 7.670)
	15-second	63.721 (± 5.371)	65.128 (± 5.603)

Table 1: Viewing behavior consistency in terms of mTC and SRM (and the standard error) under different viewing conditions.

former (*i.e.*, 7 seconds). The starting points and exploration times are combined in pairs, giving rise to four viewing conditions.

2.3 Subjective Methodology

The single stimulus continuous quality evaluation method described in the ITU-R BT 500.13 recommendation [6] is employed for our psychophysical study. Subjects need to rate the perceived quality of a 360° video on a continuous scale of [1, 5], labeled by five quality levels (“bad”, “poor”, “fair”, “good”, and “excellent”). We gather human data from 139 subjects (75 females and 64 males with ages between 17 and 26). Each video is rated by no less than 20 subjects. The absolute category rating is adopted to collect the mean opinion score (MOS) of each video, where a higher MOS means better perceived quality [7]. It is noteworthy that each video is viewed only once by one subject to ensure that user data is collected without prior knowledge of the scene.

3 ANALYSIS AND DISCUSSION

The analysis is mainly focused on comparing the impact of viewing conditions on viewing behaviors and perceived quality. We define the scanpath of a user as the [pitch, yaw] sequence from the HM data [3]. To compare multiple scanpaths, we adopt two wide-used metrics: *mean temporal correlation* (mTC) [1] and *similarity ring metric* (SRM) [3]. The higher values of mTC and SRM represent more similarity in viewing behaviors. We also consider comparing the saliency maps (*i.e.*, the heatmaps) as spatial aggregations of scanpaths for further analysis. MOSs are calculated from raw human scores using the method in [6].

3.1 How Viewing Conditions Affect Human Behaviors?

We calculate mTC and SRM under different viewing conditions, as listed in Table 1. According to the results, we find that human viewing behavior consistency is relatively low as measured *globally* by mTC for all viewing conditions. When measured more *locally* by SRM, the consistency improves. We also employ analysis of variance (ANOVA) to see whether such differences in viewing behavior consistency as measured by mTC and SRM are statistically significant. The results show the differences in consistency for different starting points and exploration times are statistically significant, as evidenced by *p*-values close to zero.

We then compute the mean PLCC values between the heatmaps from Starting Point I and Starting Point II, averaged over different videos, as shown in Fig. 2. We find that, in the first four seconds,

	Starting Point I	Starting Point II
7-second	2.551 (± 0.034)	3.059 (± 0.034)
15-second	3.119 (± 0.036)	2.562 (± 0.038)

Table 2: Perceived quality analysis in terms of MOS (and the associated standard error) under different viewing conditions.

the PLCC is lower than the initial, indicating divergent and different heatmaps, which is explained by different viewing conditions. Second, the PLCC increases over time but is still not sufficiently high to eliminate the impact of viewing conditions.

3.2 How Viewing Conditions Affect Perceived Quality?

We calculate MOS under different viewing conditions, as in Table 2. Several useful findings are worth mentioning. First, compared to 15-second exploration, 7 seconds mean fewer viewpoints of the scene to be observed, highlighting the importance of the starting point to the perceived video quality. Second, if a longer exploration time is allowed, the viewpoints close to the end are more likely to affect the overall viewing experience due to the *recency effect* [5]. This explains that for the 7-second exploration, the subjects tend to give low-quality scores when viewing from Starting Point I, where distortions appear in the initial viewpoints (see Sec. 2.2 for the definitions of the two types of starting points). If they are allowed more time, the subjects would consciously re-orient their heads to avoid viewing distorted viewpoints, and look for those with better quality, leading to an improved viewing experience. On the contrary, from Starting Point II where the distortions may not be viewed initially, the subjects are less likely to observe distortions with less allowable time, explaining the higher average MOS of the 7-second exploration.

4 CONCLUSION

We have put together the first user-generated VR video dataset that includes MOSs and viewing behavioral data. This dataset encompasses 139 users viewing 360° videos from four different conditions, resulting in a total of 40,268 opinion scores and scanpaths. We conducted a statistical analysis of various effects on viewing behaviors and perceived quality in VR, identifying viewing conditions to be crucial factors.

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